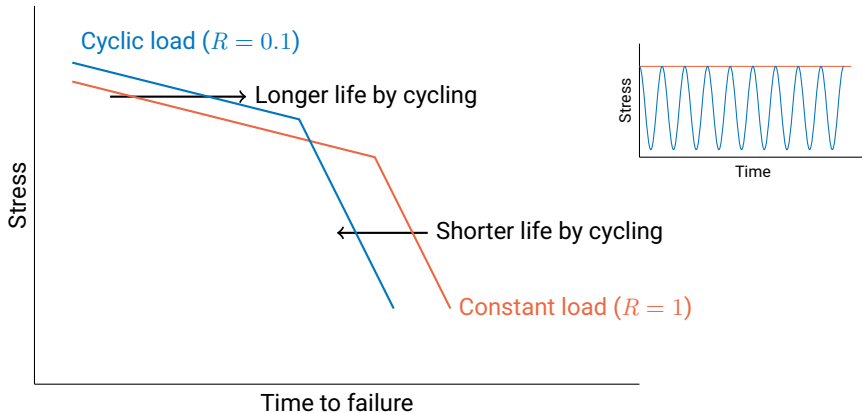


Microscale modeling of creep and fatigue in composites: viscoplasticity, cohesive damage and time homogenization

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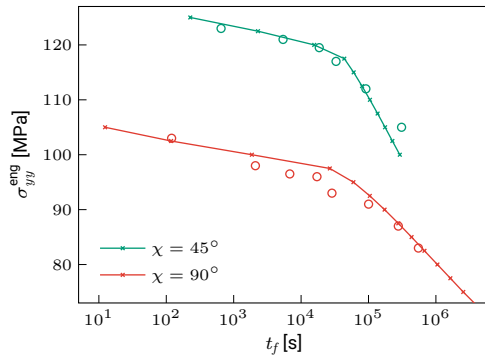
Cyclic loading has a two-fold effect on the lifetime of polymers



Are these phenomena also observable in composites?

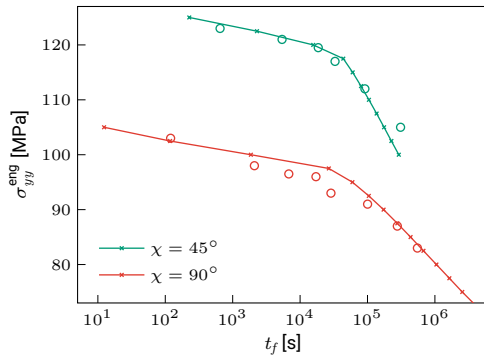
The model is calibrated on a single frequency with two different angles

Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



The model is calibrated on a single frequency with two different angles

Calibrating against experiments at 45° and 90° with $R = 0.1$ and $f = 1$ Hz



- EGP spectrum from earlier creep simulations
- Cohesive strength to match plasticity regime
 - Tensile strength σ_c for 90°
 - Shear strength τ_c for 45°
- Fatigue parameters for crack growth regime
 - Endurance limit ϵ for offset
 - S-N parameter β for slope

- Transition between regimes can be improved with recalibration of EGP spectrum
- Dependence on fracture mode in the cohesive model needs more attention

Conclusions & discussion

A unified micromechanical framework for modeling creep and fatigue in composites

- ✓ Large deformation nonlinear visco-elastic polymer model: EGP
- ✓ Cohesive zone model with cyclic damage: Dávila
- ✓ Time-homogenization for efficient time stepping with cyclic load

The model reproduces experimentally observed phenomena from off-axis tests

- Two failure regimes with different frequency dependence are distinguished in fatigue simulations
- Creep rupture ($R = 1$) is captured with the same model
- Improvement is possible in dependence on R -value and mode of fracture

[Kovačević et al (2024), *JMPS*]

